Find common multiples

Use the scaffold, show each step and ask for an adult check when marked.

Name: ______________________________ Date: ____________

1. Read the model: Find the lowest common multiple of 18 and 24. Copy or complete the first step.

2. Write the first three common multiples of 6 and 8.

3. Find the LCM of 15 and 20.

4. Miro says: Miro lists factors when the question asks for multiples. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the lowest common multiple of 18 and 24. Copy or complete the first step.
   Answer 72.

2. Write the first three common multiples of 6 and 8.
   Answer 24, 48 and 72.

3. Find the LCM of 15 and 20.
   Answer 60.

4. Miro says: Miro lists factors when the question asks for multiples. Is this correct? Explain.
   Answer Multiples continue from repeated multiplication. Factors divide the number exactly.

Find common multiples

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. Find the lowest common multiple of 18 and 24.

2. Write the first three common multiples of 6 and 8.

3. Find the LCM of 15 and 20.

4. Solve this independently and show every stage: Find the lowest common multiple of 18 and 24.

5. Use a second method or representation for: Write the first three common multiples of 6 and 8.

6. Check this result using an inverse, estimate or known fact: Find the LCM of 15 and 20.

7. Two traffic lights flash every 12 and 18 seconds. They flash together now. When next?

8. Identify and correct this misconception: Miro lists factors when the question asks for multiples.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Find the LCM of 15 and 20.

Teacher answers and checking prompts

1. Find the lowest common multiple of 18 and 24.
   Answer 72.

2. Write the first three common multiples of 6 and 8.
   Answer 24, 48 and 72.

3. Find the LCM of 15 and 20.
   Answer 60.

4. Solve this independently and show every stage: Find the lowest common multiple of 18 and 24.
   Answer 72.

5. Use a second method or representation for: Write the first three common multiples of 6 and 8.
   Answer Expected result: 24, 48 and 72. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Find the LCM of 15 and 20.
   Answer Expected result: 60. A valid check must be shown.

7. Two traffic lights flash every 12 and 18 seconds. They flash together now. When next?
   Answer After 36 seconds.

8. Identify and correct this misconception: Miro lists factors when the question asks for multiples.
   Answer Multiples continue from repeated multiplication. Factors divide the number exactly.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Find the LCM of 15 and 20.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Find common multiples

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: ___________________

1. Two traffic lights flash every 12 and 18 seconds. They flash together now. When next?

2. Prove the answer to this question, rather than only calculating it: Find the lowest common multiple of 18 and 24.

3. Solve this in two different ways and compare them: Write the first three common multiples of 6 and 8.

4. Work backwards from the answer to reconstruct the question: 60.

5. Explain why this reasoning fails: Miro lists factors when the question asks for multiples.

6. Create a new example that tests the same idea as: Find the LCM of 15 and 20.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Two traffic lights flash every 12 and 18 seconds. They flash together now. When next?
   Answer After 36 seconds.

2. Prove the answer to this question, rather than only calculating it: Find the lowest common multiple of 18 and 24.
   Answer Expected result: 72. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: Write the first three common multiples of 6 and 8.
   Answer Expected result: 24, 48 and 72. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 60.
   Answer One valid reconstruction is: Find the LCM of 15 and 20. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro lists factors when the question asks for multiples.
   Answer Multiples continue from repeated multiplication. Factors divide the number exactly.

6. Create a new example that tests the same idea as: Find the LCM of 15 and 20.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.